Day 3 Cheat Sheet: Confidence Intervals & the Bootstrap

From 'is it real?' to 'how big, give or take?'

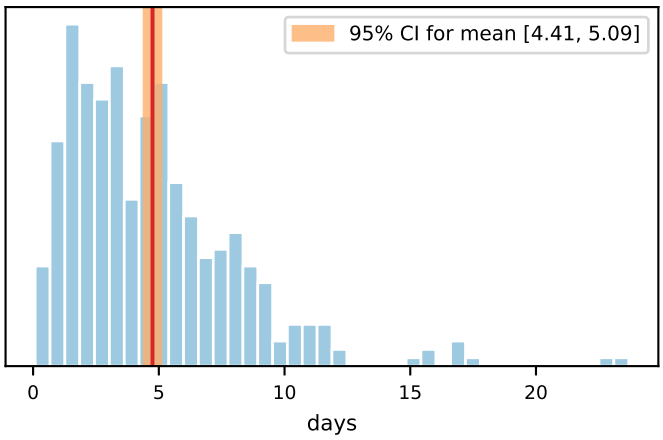
CONFIDENCE INTERVAL = estimate +/- (critical value x standard error)
- Standard error (SE) = s / sqrt(n): how much the estimate wobbles sample to sample
- 95% CI for a mean: m +/- t(0.975, n-1) x SE (t is ~1.97 when n = 400)
- 95% CI for a rate: use Wilson, not Wald, when n is small or the rate is near 0 or 1
- Quadruple n to halve the margin of error (SE shrinks with sqrt(n))

WHAT 95% MEANS: if you repeated the study many times, about 95% of the intervals would contain the true value. It is NOT "95% chance the truth is in THIS interval".

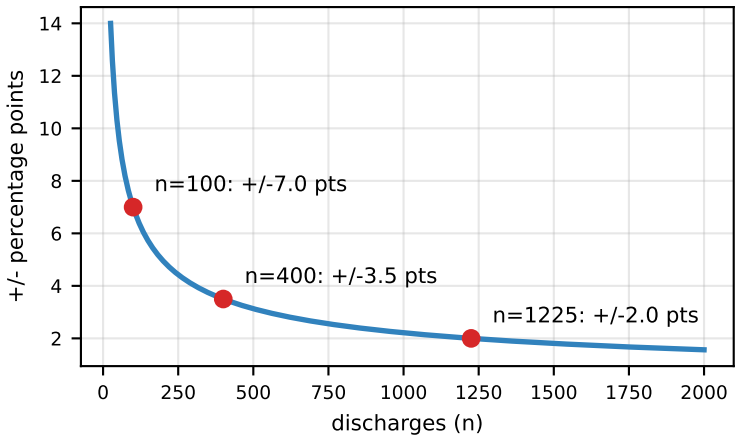
BOOTSTRAP (no formula needed, works for medians, ratios, AUC...)
1. Resample your data WITH replacement, same size n 2. Compute the statistic
3. Repeat 5,000 times 4. Take the 2.5th and 97.5th percentiles = 95% CI

LINK TO TESTS: a 95% CI for a difference that includes 0 matches p > 0.05.

Heart-failure LOS (n=400): mean 4.75 days



Margin of error for a 15% readmission rate vs. sample size



THIS DATA (simulated)
Mean LOS 4.75 days, 95% CI [4.41, 5.09] | Readmits 62/400 = 15.5%, Wilson CI [12.3%, 19.4%]
Median claim $420, bootstrap CI [$407, $432] | PPO-HMO median gap $20, CI [-$2, $50] includes 0
Sample size for +/-2 pts on a 15% rate: n = 1.96^2 x 0.15 x 0.85 / 0.02^2 = about 1,225

COMMON MISTAKES
- Reporting a p-value with no interval: stakeholders need the size of the effect, give or take
- Using the population SD formula: use s = x.std(ddof=1) for a sample
- Wald CI on a tiny clinic sample: a negative lower bound is a red flag, switch to Wilson
- Bootstrapping too few times: use at least 2,000 to 5,000 resamples

```
se = s / np.sqrt(n)                          # standard error
stats.t.interval(0.95, df=n-1, loc=m, scale=se)   # CI for a mean
stats.binomtest(k, n).proportion_ci(method='wilson')  # CI for a rate
boot = [np.median(rng.choice(x, len(x))) for _ in range(5000)]
np.percentile(boot, [2.5, 97.5])             # percentile bootstrap CI
stats.bootstrap((x,), np.median, n_resamples=5000)  # built-in (BCa)
```

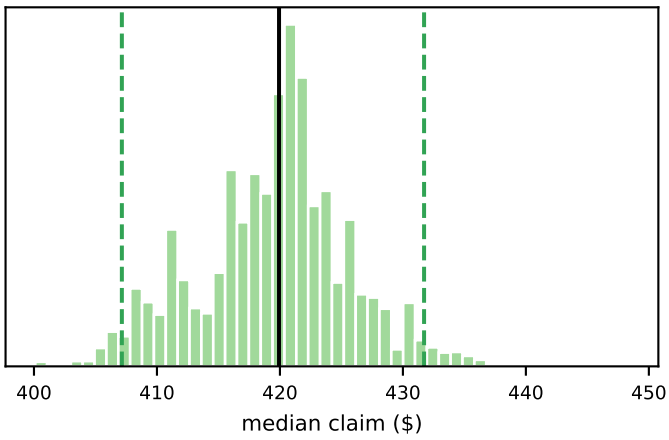
Infographic: How the Bootstrap Builds a Confidence Interval

Resampling 5,000 simulated claims, the same data as Day 2

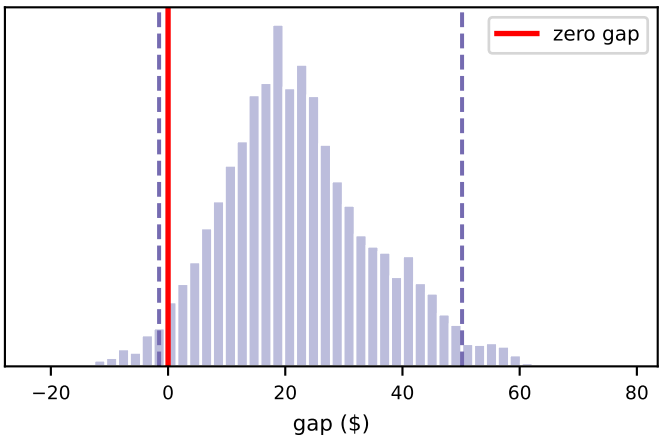
Sample — your n=5,000 claims → **Resample** — n draws WITH replacement → **Statistic** — median of the resample → **Repeat** — 5,000 times → **Percentiles** — 2.5th & 97.5th = 95% CI

MEAN LOS	READMIT RATE	MEDIAN CLAIM	MEDIAN GAP
4.75 d	15.5%	$420	$20
[4.41, 5.09]	[12.3%, 19.4%]	[$407, $432]	[-$2, $50]

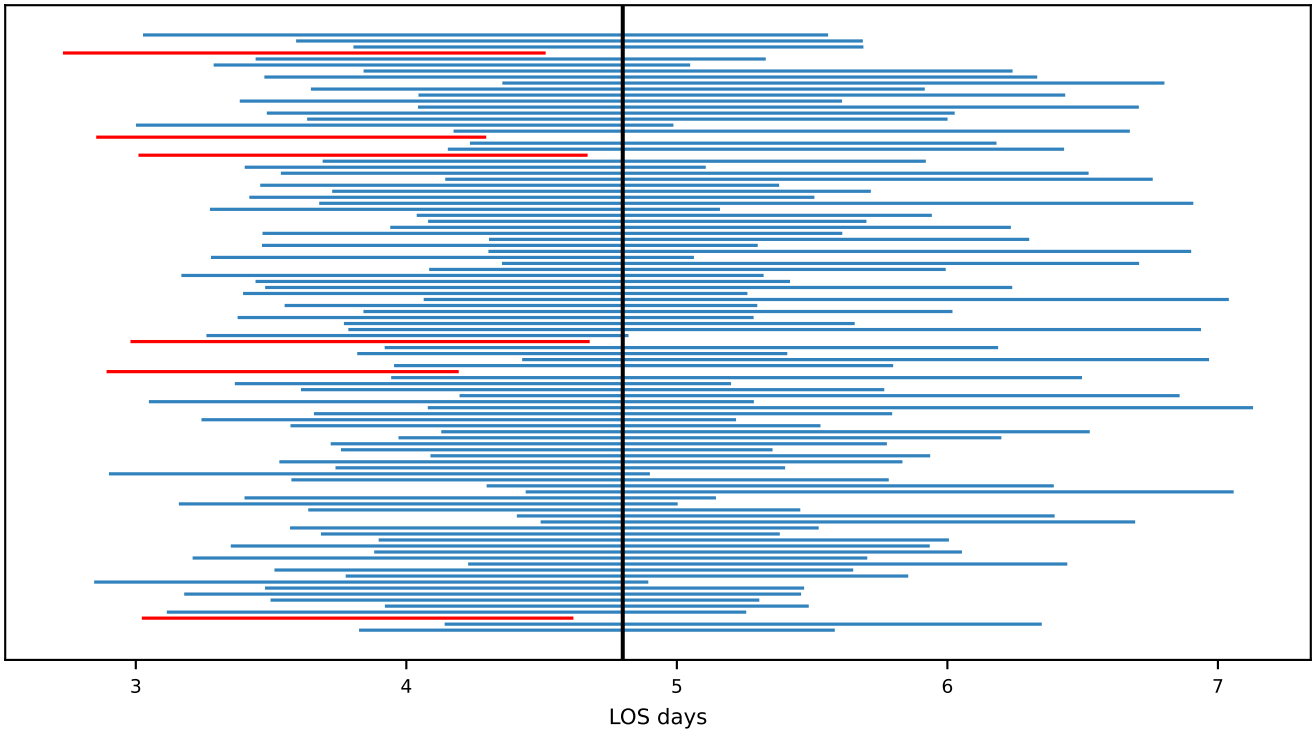
5,000 bootstrap medians: 95% CI [$407, $432]



Bootstrap PPO-HMO median gap: $20, 95% CI [-$2, $50]



100 samples of 40 patients: 94 of 100 CIs contain the true mean (4.8)



The gap's CI crosses 0, matching Day 2's permutation p of about 0.10. A CI shows the size AND the uncertainty at once.

Hands-On Worksheet: ds-genai-90-days / day03

60-90 minutes | notebook: day03/confidence_intervals_bootstrap.ipynb

[ ] mkdir day03, open VS Code, conda activate ds90 (no new packages needed)

[ ] Simulate 400 heart-failure LOS values: rng(6).gamma(2.0, 2.4, 400)

[ ] Compute mean, SE and a 95% t-interval by hand, then check with stats.t.interval

[ ] Readmissions: Wald vs. Wilson CIs for 62/400 and 2/25. Which Wald CI breaks?

[ ] Plot margin of error vs. n for a 15% rate; find n for +/-2 points

[ ] Rebuild the Day 2 claims (seed 42); bootstrap the median 5,000 times

[ ] Bootstrap the PPO-HMO median gap. Does the 95% CI include 0?

[ ] Cross-check the median CI with scipy.stats.bootstrap

[ ] Coverage: 100 samples of 40, plot 100 CIs, count how many hit 4.8

[ ] Write 3 stakeholder sentences: estimate + CI + plain-English meaning

[ ] git add day03 && git commit -m 'Day 3: confidence intervals and bootstrap' && git push

[ ] Stretch: Snowflake: mean LOS +/- 1.96 x STDDEV / SQRT(COUNT(*)) by hospital

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(6)
los = rng.gamma(shape=2.0, scale=2.4, size=400)
n, m, s = len(los), los.mean(), los.std(ddof=1)
se = s / np.sqrt(n)
lo, hi = stats.t.interval(0.95, df=n - 1, loc=m, scale=se)
print(f"{m:.2f} [{lo:.2f}, {hi:.2f}]")   # expect 4.75 [4.41, 5.09]

ci = stats.binomtest(62, 400).proportion_ci(method="wilson")
print(ci.low, ci.high)                   # expect 0.123, 0.194
```

Mean LOS CI: [_____, _____] Wilson CI for 2/25: [_____, _____]

Median gap CI: [_____, _____] Includes 0? ____ Coverage: ____ / 100

Day 3 Self-Check Quiz

1. Mean LOS is 4.75 days, 95% CI [4.41, 5.09]. Explain it to a nurse manager.

   _____

2. Does '95%' mean a 95% chance the true mean is inside THIS interval?

   _____

3. You quadruple the sample from 400 to 1,600. What happens to the CI width?

   _____

4. 2 readmissions out of 25. Why is the Wald CI a bad choice here?

   _____

5. In a bootstrap, why do you resample WITH replacement and keep size n?

   _____

6. Name two statistics with no easy CI formula where the bootstrap helps.

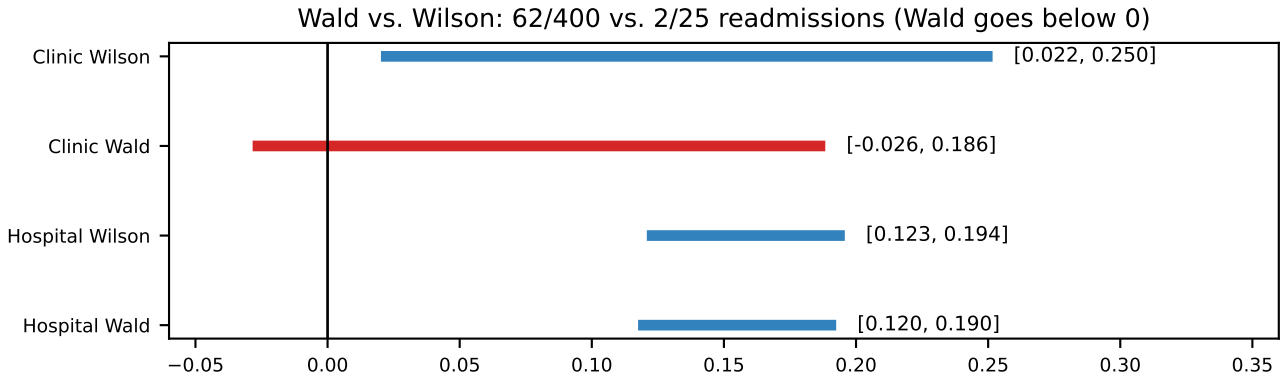
   _____

7. The PPO-HMO median gap CI is about [-$2, $50]. What does that tell you?

   _____

8. In the coverage plot, about how many of 100 CIs should miss the truth?

   _____



Answers: 1 our best estimate is 4.75 days; the true average is plausibly 4.4 to 5.1 2 No: 95% describes the method across repeats
3 it halves (SE ~ 1/sqrt(n)) 4 small n, rate near 0: it goes below 0 5 to mimic drawing new samples of the same size
6 median, percentiles, ratios, AUC 7 a gap from about zero up to $50: not clearly different from 0 (p > 0.05) 8 about 5